

National Curriculum of Pakistan 2022-2023

Assessment Framework CHEMISTRY (Theory)

Details of Content Areas/ SLOs Grade XI (11)

Domain	Content Area	SLO No.	SLO Description	Cognitive Level	Type of Assessment	Remarks	Time allocation (lectures)
A	Nature of Science in Chemistry	SLO:C-11-A-01	Describe how Al Ghazali's burning cotton thought experiment highlight the challenges of inductive reasoning examples of deductive reasoning with respect to chemistry.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	01
B	Atomic Structure	SLO: C-11-B-01	Describe that, each atomic shell and subshell are further divided into degenerate orbitals having the same energy.	Understanding	Summative	Question(s) will be asked in the annual examination.	10
		SLO: C-11-B-02	Describe protons, neutrons and electrons in terms of their relative charges and relative masses.	Understanding	Summative		
		SLO: C-11-B-03	Recognize that the terms atomic and proton number represent the same concept.	Understanding	Summative		
		SLO: C-11-B-04	Recognize the terms mass and nucleon number represent the same concept.	Understanding	Summative		
		SLO: C-11-B-05	Describe the behavior of beams of protons, neutrons and electrons moving at the same velocity in an electric field.	Understanding	Summative		
		SLO: C-11-B-06	Determine the numbers of protons, neutrons and electrons present in both atoms and ions given atomic or proton number, mass/or nucleon number and charge.	Knowledge	Summative		
		SLO: C-11-B-07	Explain the change in atomic and ionic radius across a period and down a group.	Understanding	Summative		
		SLO: C-11-B-08	Determine the electronic configuration of elements and their ions with proton numbers. (Some examples include:	Knowledge	Summative		

			a). simple configuration e.g. 2,8, b). Subshells e.g. $1s^2$, $2s^2$, $2p^6$, $3s^1$ c). Students should be able to determine both of these from the periodic table and are not required to memorize these. d) Students should understand that chemical properties of an atom are governed by valence electrons).				
		SLO: C-11-B-09	Define terms related to electronic configuration (Some examples include shells, subshells, orbitals, principal quantum number (n), ground state).	Knowledge	Summative		
		SLO: C-11-B-10	Relate Quantum Numbers to Electronic distribution of elements.	Understanding	Summative		
		SLO: C-11-B-11	Describe the number of orbitals making up s, p d and f subshells, and the number of electrons that can fill s, p d and f subshells.	Understanding	Summative		
		SLO: C-11-B-12	Apply aufbau principle, pauli exclusion principle and hunds rule to write the electronic configuration of elements	Application	Summative		
		SLO: C-11-B-13	Describe the order of increasing energy of the subshells (s,p,d and f).	Understanding	Summative		
		SLO: C-11-B-14	Describe the electronic configurations to include the number of electrons in each shell, subshell and orbital.	Understanding	Summative		
		SLO: C-11-B-15	Explain the electronic configurations in terms of energy of the electrons and inter-electron repulsion.	Understanding	Summative		
		SLO: C-11-B-16	Determine the electronic configuration of atoms and ions given the proton or electron number and charge,	Understanding	Summative		
		SLO: C-11-B-17	Illustrate the importance of electronic configurations in the development of new materials for electronic devices. (For example, semiconductors such as silicon have a specific electronic configuration that makes them ideal for use in electronic devices.)	Understanding	Summative		
		SLO: C-11-B-18	Describe the shapes of s, p and d orbitals.	Understanding	Summative		
		SLO: C-11-B-19	Describe a free radical as a species with one or more unpaired electrons.	Understanding	Summative		
		SLO: C-11-B-20	Explain that ionization energies are due to the attraction between the nucleus and the outer electron.	Understanding	Summative		
		SLO: C-11-B-21	Explain how ionization energy helps account for the trends across a period and down a group of the Periodic Table.	Understanding	Summative		

		SLO: C-11-B-22	Account for the variation in successive ionization energies of an element.	Understanding	Summative		
		SLO: C-11-B-23	Explain the factors influencing the ionization energies of elements in terms of nuclear charge, atomic/ionic radius, shielding by inner shells and subshells and spin-pair repulsion.	Understanding	Summative		
		SLO: C-11-B-24	Deduce the electronic configurations of elements using successive ionization energy data.	Application	Summative		
		SLO: C-11-B-25	Deduce the position of an element in the Periodic Table using successive ionization energy data.	Application	Summative		
		SLO: C-11-B-26	Explain how a mass spectrometer can be used to determine the relative atomic mass of an element from its isotopic composition.	Understanding	Summative		
		SLO: C-11-B-27	Perform calculations involving non-integer relative atomic masses and abundance of isotopes from given data, including mass spectra.	Application	Summative		
	Chemical Bonding	SLO: C-11-B-29	Define electronegativity as the power of an atom to attract electrons to itself.	Knowledge	Summative	Question(s) will be asked in the annual examination.	13
		SLO: C-11-B-30	Explain the factors influencing the electronegativities of elements in terms of nuclear charge, atomic radius, shielding by inner shells and subshells.	Understanding	Summative		
		SLO: C-11-B-31	Explain the trends in electronegativity across a period and down a group of the Periodic Table.	Understanding	Summative		
		SLO: C-11-B-32	Use the differences in Pauling electronegativity values to predict the formation of ionic and covalent bonds.	Application	Summative		
		SLO: C-11-B-33	Describe covalent bonding in molecules using the concept of hybridization to describe sp, sp ² and sp ³ orbitals.	Understanding	Summative		
		SLO: C-11-B-34	Use bond energy values and the concept of bond length to compare the reactivity of covalent molecules	Application	Summative		
		SLO: C-11-B-35	Describe the shapes and bond angles in molecules using VSEPR theory (including describing by sketching)	Understanding	Summative		
		SLO: C-11-B-36	Predict the shapes, and bond angles in molecules and ions.	Understanding	Summative		

		SLO: C-11-B-37	Explain hybridization and types of hybridization.	Understanding	Summative		
		SLO: C-11-B-38	Explain valence bond theory.	Understanding	Summative		
		SLO: C-11-B-39	Explain the importance of VSEPR theory in the field of drug design by discussing how the shape and bond angles of the molecules helps chemists predict their interactions in the body.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	
	Chemical Bonding	SLO: C-11-B-29	Define electronegativity as the power of an atom to attract electrons to itself.	Knowledge	Summative	Question(s) will be asked in the annual examination.	13
		SLO: C-11-B-30	Explain the factors influencing the electronegativities of elements in terms of nuclear charge, atomic radius, shielding by inner shells and subshells.	Understanding	Summative		
		SLO: C-11-B-31	Explain the trends in electronegativity across a period and down a group of the Periodic Table.	Understanding	Summative		
		SLO: C-11-B-32	Use the differences in Pauling electronegativity values to predict the formation of ionic and covalent bonds.	Application	Summative		
		SLO: C-11-B-33	Describe covalent bonding in molecules using the concept of hybridization to describe sp, sp ² and sp ³ orbitals.	Understanding	Summative		
		SLO: C-11-B-34	Use bond energy values and the concept of bond length to compare the reactivity of covalent molecules	Application	Summative		
		SLO: C-11-B-35	Describe the shapes and bond angles in molecules using VSEPR theory (including describing by sketching)	Understanding	Summative		
		SLO: C-11-B-36	Predict the shapes, and bond angles in molecules and ions.	Understanding	Summative		
		SLO: C-11-B-37	Explain hybridization and types of hybridization.	Understanding	Summative		
		SLO: C-11-B-38	Explain valence bond theory.	Understanding	Summative		
		SLO: C-11-B-39	Explain the importance of VSEPR theory in the field of drug design by discussing how the shape and bond angles of the molecules helps chemists predict their interactions in the body.	Understanding	Formative	Question(s) will not be asked in the annual	

						examination, however it will be the part of classroom teaching.	
		SLO: C-11-B-40	Explain the salient features of molecular orbital theory.	Understanding	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-B-41	Explain the paramagnetic nature of Oxygen molecule in the light of MOT.	Understanding	Summative		
		SLO: C-11-B-42	Calculate Bond order of N ₂ , O ₂ , F ₂ & He	Understanding	Summative		
		SLO: C-11-B-43	Describe the types of Van der Waals' force including a. instantaneous dipole – induced dipole (id-id) force, also called London dispersion forces b. permanent dipole – permanent dipole (pd-pd) force, including hydrogen bonding c. Hydrogen bonding as a special case of permanent dipole – permanent dipole force between molecules where hydrogen is bonded to a highly electronegative atom).	Understanding	Summative		
		SLO: C-11-B-44	Describe hydrogen bonding, limited to molecules containing N–H, O–H and H–F groups, (including ammonia, water and H–F as simple examples).	Understanding	Summative		
		SLO: C-11-B-45	Use the concept of hydrogen bonding to explain the anomalous properties of H ₂ O (ice and water).	Understanding	Summative		
		SLO: C-11-B-46	Use the concept of electronegativity to explain bond polarity and dipole moments of molecules	Understanding	Summative		
		SLO: C-11-B-47	State that, in general, ionic, covalent and metallic bonding are stronger than intermolecular forces	Understanding	Summative		
		SLO: C-11-B-48	Recognize that molecular ions/polyatomic ions can have expanded octets e.g. sulfate and nitrate	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	
		SLO: C-11-B-49	Analyze the formation of dative bond in CO, ozone and H ₃ O ⁺ ion (resonance structure not required)	Understanding	Summative	Question(s) will be asked in the annual examination.	

Stoichiometry	SLO: C-11-B-50	Express balanced chemical equations in terms of moles, representative particles, masses, and volumes of gases (at STP).	Application	Summative	Question(s) will be asked in the annual examination.	10
	SLO: C-11-B-51	Explain the concept of limiting reagents.	Understanding	Summative		
	SLO: C-11-B-52	Calculate the maximum amount of product and amount of any unreacted excess reagent.	Application	Summative		
	SLO: C-11-B-53	Calculate theoretical yield, actual yield, and percentage yield when given appropriate information.	Application	Summative		
	SLO: C-11-B-54	State the volume of one mole of a gas at STP.	Knowledge	Summative		
	SLO: C-11-B-55	Use the volume of one mole of gas at STP to solve mole volume problem.	Knowledge	Summative		
	SLO: C-11-B-56	Calculate the gram molecular mass of a gas from density measurements at STP.	Application	Summative		
	SLO: C-11-B-57	Derive measurements of mass, volume, and number of particles using moles.	Application	Summative		
	SLO: C-11-B-58	Calculate the quantities of reactants and products involved in a chemical reaction using stoichiometric principles (Some examples include calculations involving reacting masses, volumes of gasses, volumes, and concentrations of solutions, limiting reagent and excess reagent, percentage yield calculations).	Application	Summative		
	SLO: C-11-B-59	Explain, with examples, the importance of stoichiometry in the production and dosage of medicine.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of class room teaching.	
Electrochemistry	SLO: C-11-B-60	Explain the merits of photovoltaic cells as sustainable ways of meeting energy demands by making reference to the photovoltaic principle.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of class room teaching.	01

	States and phases of matter	SLO: C-11-B-61	Describe simple properties of liquids e.g., diffusion, compression, expansion, motion of molecules, spaces between them, intermolecular forces and kinetic energy based on kinetic molecular theory.	Understanding	Formative	Question(s) will not be asked in the annual examination, since already covered in previous classes. However, it will be the part of class room teaching.	08
		SLO: C-11-B-62	Describe types of intermolecular forces.	Understanding	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-B-63	Explain the strength and applications of dipole-dipole forces, hydrogen bonding and London forces.	Understanding	Summative		
		SLO: C-11-B-64	Describe physical properties of liquids such as evaporation, vapor pressure, boiling point, viscosity and surface tension.	Understanding	Summative		
		SLO: C-11-B-65	Apply the concept of hydrogen bonding to explain the properties of water (specifically high surface tension, high specific heat, low vapor pressure, high heat of vaporization, and high boiling point).	Application	Summative		
		SLO: C-11-B-66	Define molar heat of fusion and molar heat of vaporization.	Knowledge	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	
		SLO: C-11-B-67	Describe how heat of fusion and heat of vaporization affect the particles that make up matter.	Understanding	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-B-68	Outline the importance of heat of fusion in the study of glaciers and ice sheets (particularly while studying polar ice caps).	Understanding	Summative		
		SLO: C-11-B-69	Describe the physical properties of gases (including compressibility, expandability and pressure exerted by gases)	Understanding	Formative	Question(s) will not be asked in the annual examination, since already covered in previous classes. However, it will be the part of class room teaching.	
		SLO: C-11-B-70	Describe liquid crystals and give their uses in daily life.	Understanding	Formative		
		SLO: C-11-B-71	Differentiate liquid crystals from pure liquids and crystalline solids.	Understanding	Formative		
		SLO: C-11-B-72	Describe simple properties of solids e.g. compression, expansion, motion of molecules, interparticle space, intermolecular forces and kinetic energy based on kinetic molecular theory.	Understanding	Formative		

		SLO: C-11-B-73	Differentiate between amorphous and crystalline solids.	Knowledge	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-B-74	Describe properties of crystalline solids like geometrical shape, melting point, cleavage planes, habit of a crystal, crystal growth.	Understanding	Summative		
	Energetics	SLO: C-11-B-75	Describe that chemical reactions are accompanied by enthalpy changes and these changes can be exothermic (ΔH is negative) or endothermic (ΔH is positive).	Understanding	Summative	Question(s) will be asked in the annual examination.	10
		SLO: C-11-B-76	Interpret a reaction pathway diagram, in terms of the enthalpy change of the reaction and of the activation energy.	Application	Summative		
		SLO: C-11-B-77	Define terms such as standard conditions, enthalpy change, reaction, formation, combustion, neutralization.	Understanding	Summative		
		SLO: C-11-B-78	Explain that energy transfer occurs during chemical reactions because of the breaking and making of bonds.	Understanding	Summative		
		SLO: C-11-B-79	Calculate the bond energies for the enthalpy change of reaction, ΔH .	Application	Summative		
		SLO: C-11-B-80	Describe that some bond energies are exact, and some bond energies are approximate.	Understanding	Summative		
		SLO: C-11-B-81	Calculate enthalpy changes from appropriate experimental results, including the use of the relationships $q = mc\Delta T$ and $\Delta H = -mc\Delta T/n$	Application	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	
		SLO: C-11-B-82	Define terms such as enthalpy change of atomization, ΔH_{atm} , lattice energy, $\Delta H_{\text{Lattice}}$, first electron affinity, EA	Knowledge	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-B-83	Use terms such as enthalpy change of atomization, ΔH , lattice energy, ΔH , first electron affinity, EA.	Application	Summative		
		SLO: C-11-B-84	Explain the factors affecting the electron affinities of elements.	Understanding	Summative		
		SLO: C-11-B-85	Construct Born–Haber cycles for ionic solids.	Application	Summative		
		SLO: C-11-B-86	Perform calculations involving Born–Haber cycles.	Application	Summative		

		SLO: C-11-B-87	Explain the effect of ionic charge and ionic radius on the numerical magnitude of lattice energy.	Understanding	Summative		
		SLO: C-11-B-88	Apply enthalpy change with reference to hydration, and solution.	Application	Summative		
		SLO: C-11-B-89	Construct an energy cycle involving enthalpy change of solution, lattice energy and enthalpy change of hydration.	Application	Summative		
		SLO: C-11-B-90	Perform calculations involving the energy cycles.	Application	Summative		
		SLO: C-11-B-91	Explain the effect of ionic charge and ionic radius on the numerical magnitude of an enthalpy change of hydration.	Understanding	Summative		
		SLO: C-11-B-92	Define the term entropy, S, as the number of possible arrangements of the particles and their energy in a given system.	Understanding	Summative		
		SLO: C-11-B-93	Explain the sign of the entropy changes that occur during a change in state, temperature change and a reaction in which there is a change in the number of gaseous molecules.	Understanding	Summative		
		SLO: C-11-B-94	Calculate the entropy change for a reaction, ΔS , given the standard entropies, S, of the reactants and products.	Application	Summative		
		SLO: C-11-B-95	Explain the concept of heat as a form of energy.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	
		SLO: C-11-B-96	Explain the relationship between temperature and kinetic energy of particles.	Understanding	Formative		
		SLO: C-11-B-97	State that total energy is conserved in chemical reaction.	Knowledge	Formative		
		SLO: C-11-B-98	Explain the concept of standard conditions and standard states in measuring energy changes.	Understanding	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-B-99	Explain Hess's Law.	Understanding	Summative		
		SLO: C-11-B-100	Apply Hess's Law to calculate enthalpy changes in a reaction carried out in multiple steps.	Application	Summative		
		SLO: C-11-B-101	Explain the relationship between bond formation energy, and bond breaking energy.	Understanding	Summative		
		SLO: C-11-B-102	Explain Gibbs free energy.	Understanding	Summative		

		SLO: C-11-B-103	Apply the concept of Gibbs free energy to solve problems.	Application	Formative	Question(s) will not be asked in the annual examination, since already covered in previous classes. However, it will be the part of class room teaching.	
		SLO: C-11-B-104	Outline how enthalpy change relates to the calorie content of the food we eat.	Application	Summative	Question(s) will be asked in the annual examination.	
Chemical Kinetics	SLO: C-11-B-105	Explain the rate of reaction and rate constant.	Understanding	Summative	Question(s) will be asked in the annual examination.	07	
	SLO: C-11-B-106	Use experimental data to calculate the rate of a reaction.	Application	Summative			
	SLO: C-11-B-107	Explain the concept of activation energy and its role in chemical reactions.	Understanding	Summative			
	SLO: C-11-B-108	Use the Boltzmann distribution curve to explain the effect of temperature on the rate of a reaction.	Understanding	Summative			
	SLO: C-11-B-109	Explain the concept of catalyst and how they increase the rate of a reaction by lowering the activation energy.	Understanding	Summative			
	SLO: C-11-B-110	Interpret reaction pathway diagrams, including in the presence and absence of catalysts.	Application	Summative			
	SLO: C-11-B-111	Explain the relationship between Gibbs free energy change, ΔG , and the feasibility of a reaction.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of class room teaching.		
	SLO: C-11-B-112	Use rate equations, including orders of reaction and rate constants.	Application	Summative	Question(s) will be asked in the annual examination.		
	SLO: C-11-B-113	Calculate the numerical value of a rate constant using the initial rates and half-life method.	Application	Summative			
	SLO: C-11-B-114	Suggest a reaction mechanism that is consistent with a given rate equation and rate-determining step.	Understanding	Summative			

		SLO: C-11-B-115	Describe the effect of temperature change on the rate constant and rate of a reaction.	Understanding	Summative		
	Chemical Equilibrium	SLO: C-11-B-116	Describe what is meant by a reversible reaction and dynamic equilibrium in terms of the rate of forward and reverse reactions being equal and the concentration of reactants and products remaining constant.	Understanding	Summative	Question(s) will be asked in the annual examination.	10
		SLO: C-11-B-117	Define dynamic equilibrium between two physical states.	Knowledge	Summative		
		SLO: C-11-B-118	State the necessary conditions for equilibrium and the ways that equilibrium can be recognized.	Knowledge	Summative		
		SLO: C-11-B-119	Describe the microscopic events that occur when a chemical system is in equilibrium Define with examples.	Understanding	Summative		
		SLO: C-11-B-120	Deduce the equilibrium constant expression K_c from an equation for homogeneous reaction.	Understanding	Summative		
		SLO: C-11-B-121	Determine the relationship between different equilibrium constants (K_c) for the same reaction at the same temperature.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of class room teaching.	
		SLO: C-11-B-122	Write the equilibrium expression for a given chemical reaction in terms of concentration, K_c , partial pressure, number of moles and mole fraction.	Application	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-B-123	Differentiate between Microscopic and Macroscopic events in a chemical reaction.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of class room teaching.	
		SLO: C-11-B-124	Propose microscopic events that account for observed macroscopic changes that take place during a shift in equilibrium.	Understanding	Formative		
			SLO: C-11-B-125	Determine if the equilibrium constant will increase or decrease when temperature is changed, given the equation for the reaction.	Understanding	Summative	Question(s) will be asked in the annual examination.
		SLO: C-11-B-126	State Le Chatelier's Principle and be able to apply it to systems in equilibrium with changes in concentration, pressure, temperature, or the addition of catalyst.	Knowledge	Summative		

		SLO: C-11-B-127	Explain industrial applications of Le Chatelier's Principle using Haber's process and the Contact Process as an example.	Understanding	Summative		
		SLO: C-11-B-128	Discuss the industrial applications of chemical equilibria and how it can be used to optimize chemical reactions to maximize yields and minimize waste products.	Understanding	Summative		
		SLO: C-11-B-129	Use the concept of hydrolysis to explain why aqueous solutions of some salts are acidic or basic.	Application	Summative		
	Acids, Bases and Salts	SLO: C-11-B-130	Define conjugate acid-base pair.	Understanding	Summative	Question(s) will be asked in the annual examination.	08
		SLO: C-11-B-131	Identify conjugate acid-base pairs in reactions.	Understanding	Summative		
		SLO: C-11-B-132	Apply the concept of conjugate acid and conjugate base on salt hydrolysis.	Application	Summative		
		SLO: C-11-B-133	Define mathematically the terms pH, K_a , pK_a and K_w and use them in calculations (K_b and the equation $K_w = K_a \times K_b$ will not be tested).	Application	Summative		
		SLO: C-11-B-134	Calculate $[H^+]_{(aq)}$ and pH values for (a) strong acids (b) strong alkalis (c) weak acids (d) weak alkalis.	Application	Summative		
		SLO: C-11-B-135	Distinguish that Lewis acids accept lone pair, and Lewis bases donate lone pair to make a coordinate covalent bond.	Understanding	Summative		
		SLO: C-11-B-136	Calculate the pH of buffer solutions in given appropriate data.	Application	Summative		
		SLO: C-11-B-137	Demonstrate the ability to comprehend and effectively apply the concept of solubility product. (K_{sp})	Application	Summative		
		SLO: C-11-B-138	Construct an expression for K_{sp} .	Application	Summative		
		SLO: C-11-B-139	Calculate K_{sp} from concentrations and vice versa.	Application	Summative		
		SLO: C-11-B-140	Apply the concept of the common ion effect to describe why the solubility of a substance changes when it is dissolved in a solution containing a common ion.	Application	Summative		
		SLO: C-11-B-141	Perform calculations using K_{sp} values and concentration of a common ion.	Application	Summative		
		SLO: C-11-B-142		Understanding	Summative		

		SLO: C-11-B-143 SLO: C-11-B-144	Use the concept of hydrolysis to explain why aqueous solutions of some salts are acidic or basic. Calculate the $[H_3O^+]$ given the K_a and molar concentration of weak acid. Calculate concentrations of ions of slightly soluble salts.	Application Application	Summative Summative		
		SLO: C-11-B-145 SLO: C-11-B-146	Perform acid-base titrations to calculate molarity and strength of given sample solutions. Select suitable indicators for acid-alkali titrations, given appropriate data (pKa values will not be used).	Application Understanding	Summative for Practical Based Assessment (PBA). Summative for PBA	Lab Work, Question(s) will be asked in PBA.	

C	Periodic Table	SLO: C-11-C-01	Explain the arrangement of elements in the periodic table.	Understanding	Summative	Question(s) will be asked in the annual examination.	10
		SLO: C-11-C-02	Explain that the periodic table is arranged into four blocks associated with the four sublevels—s, p, d, and f.	Understanding	Summative		
		SLO: C-11-C-03	Recognize that the period number (n) is the outer energy level that is occupied by electrons.	Understanding	Summative		
		SLO: C-11-C-04	State that the number of the principal energy level and the number of the valence electrons in an atom can be deduced from its position on the periodic table.	Knowledge	Summative		
		SLO: C-11-C-05	Identify the positions of metals, nonmetals and metalloids in the periodic table.	Understanding	Summative		
		SLO: C-11-C-06	Explain that vertical and horizontal trends in the periodic table exist for atomic radius, ionic radius, ionization energy, electron affinity and electronegativity.	Understanding	Summative		
		SLO: C-11-C-07	Recognize that trends in metallic and non-metallic behavior are due to the trends in valence electrons.	Understanding	Summative		
		SLO: C-11-C-08	Deduce the electron configuration of an atom from the element's position on the periodic table, and vice versa (based on s,p,d and f subshells).	Understanding	Summative		

		SLO: C-11-C-09	Write equations for, the reactions of Na and Mg with oxygen, chlorine and water.	Application	Summative		
		SLO: C-11-C-10	Explain the variation in the oxidation number of the oxides and chlorides (NaCl, MgCl ₂ in terms of their outer shell (valence shell) electrons.	Understanding	Summative		
		SLO: C-11-C-11	Describe (including writing equations for) the reactions, if any, of the oxides (acidic and basic) with water (including the likely pHs of the solutions obtained).	Understanding	Summative		
		SLO: C-11-C-12	Explain with the help of equations for, the acid / base behavior of the oxides and the hydroxides NaOH, Mg(OH) ₂ including, where relevant, amphoteric behavior in reactions with acids and bases (sodium hydroxide only).	Understanding	Summative		
		SLO: C-11-C-13	Explain with equations for, the reactions of the chlorides with water including the likely pHs of the solutions obtained.	Understanding	Summative		
		SLO: C-11-C-14	Explain the variations and trends in terms of bonding and electronegativity.	Understanding	Summative		
		SLO: C-11-C-15	Suggest the types of chemical bonding present in the chlorides and oxides from observations of their chemical and physical properties.	Understanding	Summative		
		SLO: C-11-C-16	predict the characteristic properties of an element in a given group by using knowledge of chemical periodicity	Application	Summative		
		SLO: C-11-C-17	Deduce the nature, possible position in the Periodic Table and identity of unknown elements from given information about physical and chemical properties.	Understanding	Summative		
		SLO: C-11-C-18	Explain the trends in the ionization energies and electron affinities of the Group 1 and Group 17 elements.	Understanding	Summative		
	Group 17	SLO: C-11-C-19	Describe the colors and trend in volatility of chlorine, bromine and iodine.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	04
		SLO: C-11-C-20	Describe the trend in bond strength of halogen molecules.	Understanding	Formative		
		SLO: C-11-C-21	Interpret the volatility of the elements in terms of instantaneous dipole-induced dipole forces.	Understanding	Formative		

		SLO: C-11-C-22	Describe the relative reactivity of the halogen elements as oxidizing agents.	Understanding	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-C-23	Describe the reactions of the elements with hydrogen and explain their relative reactivity in these reactions.	Understanding	Summative		
		SLO: C-11-C-24	Describe the relative thermal stabilities of the hydrogen halides and explain these in terms of bond strengths.	Understanding	Summative		
		SLO: C-11-C-25	Describe the relative reactivity of halide ions as reducing agents.	Understanding	Summative		
		SLO: C-11-C-26	Explain the reactions of halide ions with aqueous silver ions and concentrated sulfuric acid.	Understanding	Summative		
		SLO: C-11-C-27	Describe the reaction of halides with aqueous silver ions followed by aqueous ammonia.	Understanding	Summative for Practical Based Assessment (PBA)	Lab Work, Question(s) will be asked in PBA.	
		SLO: C-11-C-28	Interpret the reaction of chlorine with cold and hot aqueous sodium hydroxide as disproportionation reactions.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	
		SLO: C-11-C-29	Explain the use of chlorine in water purification, including the production of the active species HOCl and ClO ⁻ which kill bacteria.	Understanding	Summative	Question(s) will be asked in the annual examination.	

	Nitrogen and Sulphur	SLO: C-11-C-30	Explain the lack of reactivity of nitrogen due to its triple bond strength and lack of polarity.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	03
		SLO: C-11-C-31	Describe the basicity of ammonia using the Brønsted–Lowry theory.	Understanding	Formative		
		SLO: C-11-C-32	Identify the structure of the ammonium ion and explain how it is formed by an acid-base reaction.	Understanding	Formative		
		SLO: C-11-C-33	Describe how ammonia can be displaced from ammonium salts through acid-base reactions.	Understanding	Formative		
		SLO: C-11-C-34	Describe the natural and man-made occurrences of oxides of nitrogen and their catalytic removal from exhaust gases of internal combustion engines.	Understanding	Formative		
		SLO: C-11-C-35	Explain the role of NO and NO ₂ in the formation of photochemical smog, specifically in the reaction with unburned hydrocarbons to form Peroxyacetyl nitrate (PAN).	Understanding	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-C-36	Differentiate between nitrification and denitrification.	Knowledge	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	
		SLO: C-11-C-37	Explain the lack of reactivity of sulfur, with reference to its bonding and stability of its compounds.	Understanding	Formative		
		SLO: C-11-C-38	Describe the different oxidation states of sulfur and their relative stability.	Understanding	Formative		
		SLO: C-11-C-39	Describe the properties and uses of sulfuric acid, including its production and industrial applications.	Understanding	Formative		
		SLO: C-11-C-40	Describe the chemical reactions and processes involving sulfur, such as combustion and oxidation.	Understanding	Formative		
	SLO: C-11-C-41	Explain the uses of sulfur compounds in industry and everyday life, such as in fertilizers, gunpowder and rubber, and in the Synthetic organic chemistry, including the synthesis of dyes, drugs and fragrances.	Understanding	Summative	Question(s) will be asked in the annual examination.		
D	Environmental Chemistry Air	SLO: C-11-D-01	Identify the properties and composition of the atmosphere. (Include the concepts of 4 layers of atmosphere and their composition).	Understanding	Formative	Question(s) will not be asked in the annual examination, since already covered in previous classes. However, it will be	03

						the part of class room teaching.	
		SLO: C-11-D-02	Describe the factors that affect air quality.	Understanding	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-D-03	Describe the sources and understand the effects of air pollution, (This can include both natural and human-caused pollutants including O ₃ , Pb, Hg, Polycyclic aromatic hydrocarbons, Persistent organic pollutants, Greenhouse gases (such as CO ₂ , CH ₄ and N ₂ O), CFCs and other ozone-depleting substances, Volatile organic compounds, Heavy metals (such as Pb, Hg, and Cd).	Understanding	Formative	Question(s) will not be asked in the annual examination, since already covered in previous classes. However, it will be the part of class room teaching.	
		SLO: C-11-D-04	Familiarize with use of the methods and techniques to measure and monitor air quality.	Understanding	Formative		
		SLO: C-11-D-05	Describe the impact of human activities on the atmosphere, including the effects of burning fossil fuels and deforestation.	Understanding	Formative		
		SLO: C-11-D-06	Identify the chemical reactions and processes that occur in the atmosphere (some examples include the formation of smog and acid rain).	Understanding	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-D-07	Identify laws and regulations related to air quality and the measures used to control air pollution.	Understanding	Formative	Question(s) will not be asked in the annual examination, since already covered in previous classes. However, it will be the part of class room teaching.	
		SLO: C-11-D-08	Analyze data and interpret air quality measurements and trends.	Understanding	Formative		
		SLO: C-11-D-09	Explain the link between air quality and human health.	Understanding	Formative		
		SLO: C-11-D-10	Evaluate the potential health risks associated with air pollution.	Understanding	Formative		
		SLO: C-11-D-11	Explain the technologies and strategies used to reduce air pollution and improve air quality, such as emissions control and renewable energy sources.	Understanding	Formative		
		SLO: C-11-D-12	Design experiments and collect data to test hypotheses about air quality	Application	Formative		
		SLO: C-11-D-13	Identify the global scale problems of air pollution, such as global warming and the greenhouse effect.	Understanding	Formative		

		SLO: C-11-D-14	Analyze the economic, social and political issues related to air pollution and air quality management and demonstrate through answers.	Understanding	Formative		
	Environmental Chemistry-Water	SLO: C-11-D-15	Identify different types of water pollution, (some examples include point source and nonpoint source pollution)	Understanding	Formative	Question(s) will not be asked in the annual examination, since already covered in previous classes. However, it will be the part of class room teaching.	03
		SLO: C-11-D-16	Identify common water pollutants (Some examples include oil, pesticides, and heavy metals)	Understanding	Formative		
		SLO: C-11-D-17	Identify and explain the sources and effects of water pollution on human health and the environment.	Understanding	Formative		
		SLO: C-11-D-18	Identify and explain water treatment methods and technologies, such as filtration and purification.	Understanding	Formative		
		SLO: C-11-D-19	Explain the laws and regulations related to water pollution and conservation.	Understanding	Formative		
		SLO: C-11-D-20	Evaluate the impact of human activities on water resources, such as agriculture and industrial processes.	Understanding	Formative		
		SLO: C-11-D-21	Explain conservation and management strategies for protecting and preserving water resources.	Understanding	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-D-22	Explain the chemical properties of water and how they relate to water quality and pollution.	Understanding	Formative	Question(s) will not be asked in the annual examination, since already covered in previous classes. However, it will be the part of class room teaching.	

E	Organic Chemistry	SLO: C-11-E-01	Recognize that hydrocarbons are compounds made up of C and H atoms only.	Understanding	Summative	Question(s) will be asked in the annual examination.	11
		SLO: C-11-E-02	Recognize that alkanes are simple hydrocarbons with no functional group.	Understanding	Summative		
		SLO: C-11-E-03	Recognize that compounds contain a functional group which dictates their physical and chemical properties.	Understanding	Summative		
		SLO: C-11-E-04	Interpret the general, structural, displayed, and skeletal formulae of the classes of compounds.	Application	Summative		
		SLO: C-11-E-05	Describe the use of systematic nomenclature of simple aliphatic organic molecules with functional groups.	Understanding	Summative		
		SLO: C-11-E-06	Deduce the molecular and/or empirical formula of a compound, given its structural, displayed, or skeletal formula.	Application	Summative		
		SLO: C-11-E-07	Describe terminology associated with the types of organic compounds and reactions (Some examples include: homologous series, saturated and unsaturated, homolytic and heterolytic fission, free radical, initiation, propagation, termination, nucleophile, electrophile, nucleophilic, electrophilic, addition, substitution, elimination, hydrolysis, condensation, oxidation and reduction)	Understanding	Summative		
		SLO: C-11-E-08	Define catenation and explain its importance in organic chemistry.	Knowledge	summative		
		SLO: C-11-E-09	Describe terminology associated with types of organic mechanisms (Some examples include: free-radical substitution, electrophilic addition, nucleophilic substitution, nucleophilic addition)	Understanding	Summative		
		SLO: C-11-E-10	Draw the mechanism of a chemical reaction using curly arrows to represent the movement of a pair of electrons in at least three different types of reactions, including nucleophilic substitution, electrophilic addition, and elimination reactions.	Understanding	Summative		
		SLO: C-11-E-11	Apply the term 'planar' when describing the arrangement of atoms in organic molecules.	Application	Summative		
		SLO: C-11-E-12	Describe structural isomerism (in the context of organic molecules) and its division into chain, positional, functional group isomerism metamerism and tautomerism.	Understanding	Summative		

Hydrocarbons	SLO: C-11-E-13	Classify hydrocarbons as aliphatic and aromatic.	Understanding	Summative	Question(s) will be asked in the annual examination.	10
	SLO: C-11-E-14	Describe nomenclature of alkanes and cycloalkanes.	Understanding	Summative		
	SLO: C-11-E-15	Explain the shapes of alkanes and cycloalkanes exemplified by ethane and cyclopropane.	Understanding	Summative		
	SLO: C-11-E-16	Explain unreactive nature of alkanes towards polar reagents.	Understanding	Summative		
	SLO: C-11-E-17	Define homolytic and heterolytic fission, free radical initiation, propagation, and termination.	Understanding	Summative		
	SLO: C-11-E-18	Describe the mechanism of free radical substitution in alkanes exemplified by methane and ethane.	Understanding	Summative		
	SLO: C-11-E-19	Identify organic redox reactions.	Understanding	Summative		
	SLO: C-11-E-20	Explain the nomenclature of alkenes.	Understanding	Summative		
	SLO: C-11-E-21	Explain shape of ethene molecule in terms of structure and bonding..	Understanding	Summative		
	SLO: C-11-E-22	Describe the structure and reactivity of alkenes as exemplified by ethene.	Understanding	Summative		
	SLO: C-11-E-23	Explain with suitable examples the terms isomerism, stereoisomerism and structural isomerism.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be the part of classroom teaching.	
	SLO: C-11-E-24	Explain dehydration of alcohols and dehydrohalogenation of RX for the preparation of ethene.	Understanding	Summative	Question(s) will be asked in the annual examination.	
	SLO: C-11-E-25	Describe the chemistry of alkenes by the following reactions of ethene: hydrogenation, hydrohalogenation, hydration, halogenation, halohydration, epoxidation, ozonolysis, polymerization.	Understanding	Summative		
SLO: C-11-E-26	Explain the concept of conjugation in alkenes having alternate double	Understanding	Summative			

		SLO: C-11-E-27	bonds. Use the IUPAC naming system for alkenes.	Understanding	Summative		
		SLO: C-11-E-28	Describe the mechanism of electrophilic addition in alkenes, using bromine / ethene and hydrogen bromide /propene as examples.	Understanding	Summative		
		SLO: C-11-E-29	Explain the inductive effects of alkyl groups on the stability of primary, secondary and tertiary cations formed during electrophilic addition (this should be used to explain Markovnikov addition).	Understanding	Summative		
	Halogeno alkanes	SLO: C-11-E-30	Classify halogenoalkanes based on the type of halogen atom and its position in the carbon chain, and explain how the molecular structure affects their reactivity.	Understanding	Summative	Question(s) will be asked in the annual examination.	05
		SLO: C-11-E-31	Explain the organic functional groups involved in a simple halogenoalkane synthesis, and predict the major product(s) based on the reaction conditions. (this includes: a) the free-radical substitution of alkanes by Cl or Br in the presence of ultraviolet light, as exemplified by the reactions of ethane. b) Electrophilic addition of an alkene with a halogen, X ₂ , or hydrogen halide, HX(g), at room temperature. c) substitution of an alcohol, e.g. by reaction with HX or KBr with H ₂ SO ₄ or H ₃ PO ₄ ; or with PCl ₃ and heat; or with PCl ₅ ; or with SOCl ₂).	Understanding	Summative		
		SLO: C-11-E-32	Describe nucleophilic substitution reactions (specifically: a. the reaction with NaOH(aq) and heat to produce an alcohol b. the reaction with KCN in ethanol and heat to produce a nitrile. c. the reaction with NH ₃ in ethanol heated under pressure to produce an amine. d. the reaction with aqueous silver nitrate in ethanol as a method of identifying the halogen present as exemplified by bromoethane).	Understanding	Summative		
		SLO: C-11-E-33	Describe the elimination reaction with NaOH in ethanol and heat to produce an alkene as exemplified by bromoethane.	Understanding	Summative		

Hydroxy Compounds	SLO: C-11-E-34	State the reactions (reagents and conditions) by which alcohols can be produced: (a) electrophilic addition of steam to an alkene, $\text{H}_2\text{O}(\text{g})$ and H_3PO_4 catalyst (b) reaction of alkenes with cold dilute acidified potassium manganate(VII) to form a diol (c) substitution of a halogenoalkane using $\text{NaOH}(\text{aq})$ and heat (d) reduction of an aldehyde or ketone using NaBH_4 or LiAlH_4 (e) reduction of a carboxylic acid using LiAlH_4 (f) hydrolysis of an ester using dilute acid or dilute alkali and heat describe the reaction with oxygen (combustion) of organic hydroxy compounds.	Knowledge	Summative	Question(s) will be asked in the annual examination. Question(s) will be asked in the annual examination.	07
	SLO: C-11-E-35	Describe substitution to halogenoalkanes, e.g. by reaction with HX or KBr with H_2SO_4 or H_3PO_4 ; or with PCl_3 and heat; or with PCl_5 ; or with SOCl_2 .	Understanding	Summative		
	SLO: C-11-E-36	Describe the reaction of hydroxy organic compounds with $\text{Na}(\text{s})$.	Understanding	Summative		
	SLO: C-11-E-37	Describe the oxidation with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 to: carbonyl compounds by distillation, carboxylic acids by refluxing (primary alcohols give aldehydes which can be further oxidized to carboxylic acids, secondary alcohols give ketones, tertiary alcohols cannot be oxidized).	Understanding	Summative		
	SLO: C-11-E-38	Describe the dehydration of alcohols to alkenes by using a heated catalyst, e.g. Al_2O_3 or a concentrated acid.	Understanding	Summative		
	SLO: C-11-E-39	Describe the formation of esters by reaction with carboxylic acids and concentrated H_2SO_4 or H_3PO_4 as catalyst as exemplified by ethanol.	Understanding	Summative		
	SLO: C-11-E-40	Classify alcohols as primary, secondary and tertiary alcohols, to include examples with more than one alcohol group.	Understanding	Summative		
	SLO: C-11-E-41	State characteristic distinguishing reactions, e.g. mild oxidation with acidified $\text{K}_2\text{Cr}_2\text{O}_7$, colour change from orange to green	Understanding	Summative		
	SLO: C-11-E-42	Deduce the presence of a $\text{CH}_3\text{CH}(\text{OH})-$ group in an alcohol, $\text{CH}_3\text{CH}(\text{OH})-\text{R}$, from its reaction with alkaline $\text{I}_2(\text{aq})$ to form a yellow precipitate of tri-iodomethane and an ion, RCO_2^-	Application	Summative		
	SLO: C-11-E-43	Explain the acidity of alcohols compared with water	Understanding	Summative		
Carbonyl Compounds	SLO: C-11-E-44	State the reactions (reagents and conditions) by which aldehydes and ketones can be produced: a. the oxidation of primary alcohols using acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or acidified KMnO_4 and distillation to produce	Understanding	Summative	Question(s) will be asked in the	08

			aldehydes b. the oxidation of secondary alcohols using acidified $K_2Cr_2O_7$ or acidified $KMnO_4$ and distillation to produce ketones			annual examination.	
		SLO: C-11-E-45	Describe: a. the reduction of aldehydes and ketones, using $NaBH_4$ or $LiAlH_4$ to produce alcohols b. the reaction of aldehydes and ketones with HCN , KCN as catalyst, and heat to produce hydroxynitriles exemplified by ethanal and propanone	Understanding	Summative		
		SLO: C-11-E-46	Describe the mechanism of the nucleophilic addition reactions of hydrogen cyanide with aldehydes and ketones.	Understanding	Summative		
		SLO: C-11-E-47	Describe the use of 2,4-dinitrophenylhydrazine (2,4-DNPH reagent) to detect the presence of carbonyl compounds.	Understanding	Summative		
		SLO: C-11-E-48	Deduce the nature (aldehyde or ketone) of an unknown carbonyl compound from the results of simple tests (Fehling's and Tollens' reagents; ease of oxidation)	Understanding	Summative		
		SLO: C-11-E-49	Deduce the presence of a $CH_3CO -$ group in an aldehyde or ketone, CH_3COR , from its reaction with alkaline $I_2(aq)$ to form a yellow precipitate of tri-iodomethane and an ion, RCO_2^- .	Application	Summative		
		SLO: C-11-E-50	Recall the reactions by which carboxylic acids can be produced: a) oxidation of primary alcohols and aldehydes with acidified $K_2Cr_2O_7$ or acidified $KMnO_4$ and refluxing b) hydrolysis of nitriles with dilute acid or dilute alkali followed by acidification c) hydrolysis of esters with dilute acid or dilute alkali and heat followed by acidification	Knowledge	Summative		
		SLO: C-11-E-51	Describe a. the redox reaction with reactive metals to produce a salt and $H_2(g)$ b. the neutralization reaction with alkalis to produce a salt and H_2O c. the acid-base reaction with carbonates to produce a salt and $H_2O(l)$ and $CO_2(g)$ d. esterification with alcohols with concentrated H_2SO_4 as catalyst e. reduction by $LiAlH_4$ to form a primary alcohol.	Understanding	Summative		
		SLO: C-11-E-52	Recall the reaction (reagents and conditions) by which esters can be produced: the condensation reaction between an alcohol and a carboxylic acid with concentrated H_2SO_4 as catalyst.	Knowledge	Summative		
		SLO: C-11-E-53	Describe the hydrolysis of esters by dilute acid and by dilute alkali and heat	Understanding	Summative		

	Nitrogen Compounds-Amines	SLO: C-11-E-54	Define primary and secondary amines, and explain their basic properties and reactivity.	Understanding	Summative	Question(s) will be asked in the annual examination.	03
		SLO: C-11-E-55	Identify the differences between primary and secondary amines in terms of their structure and chemical properties.	Understanding	Summative		
		SLO: C-11-E-56	Describe the preparation methods of primary and secondary amines, including nucleophilic substitution reactions and reduction of nitro compounds.	Understanding	Summative		
		SLO: C-11-E-57	Explain the properties and reactivity of phenylamine and azo compounds, including their use as dyes and pigments.	Understanding	Summative		
	Organic Synthesis	SLO: C-11-E-58	Explain the concept of organic synthesis and functional group interconversions.	Understanding	Summative	Question(s) will be asked in the annual examination.	03
		SLO: C-11-E-59	Identify organic functional groups using the reactions in this progression grid.	Understanding	Summative		
		SLO: C-11-E-60	Predict properties and reactions of organic molecules based on functional group presence.	Application	Summative		
		SLO: C-11-E-61	Devise multi-step synthetic routes for preparing organic molecules.	Understanding	Summative		
		SLO: C-11-E-62	Analyze a given synthetic route in terms of type of reaction and reagents used for each step of it, and possible by products.	Understanding	Summative		
		SLO: C-11-E-63	Explain the concept of retrosynthesis and its application in organic synthesis.	Understanding	Formative	Question(s) will not be asked from 63, it needs more explanation. Could be possible in grade 12	

F	Energy	SLO: C-11-F-01	Differentiate between the difference between petrochemical and chemicals derived from them,	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be part of classroom teaching.	02
		SLO: C-11-F-02	Identify the various raw materials for the petrochemical industry.	Understanding	Formative		
		SLO: C-11-F-03	Explain the process of fractional distillation and refining of petroleum and identify the important fractions.	Understanding	Formative		
		SLO: C-11-F-04	Describe the basic building block processes in petrochemical technology and explain the petrochemical process technology.	Understanding	Formative		
		SLO: C-11-F-05	List some major petrochemicals and understand the importance of petrochemicals in the modern world.	Understanding	Formative		
		SLO: C-11-F-06	Distinguish between energy density and specific energy of different energy sources, and explain the efficiency of energy transfer	Understanding	Formative		
		SLO: C-11-F-07	Explain the formation, properties, and uses of fossil fuels, and the importance of fossil fuels in the modern world.	Understanding	Summative	Question(s) will be asked in the annual examination.	
		SLO: C-11-F-08	Explain the mechanism and importance of nuclear fusion and fission and explain the importance of nuclear energy in the modern world.	Understanding	Formative	Question(s) will not be asked in the annual examination, however it will be part of classroom teaching.	
		SLO: C-11-F-09	Explain the importance and mechanism of solar energy and its importance as a source of renewable energy in the modern world.	Understanding	Formative		
		SLO: C-11-F-10	Explain the environmental impact of energy consumption, particularly in relation to global warming and its relation to the importance of reducing carbon footprint and moving towards sustainable energy sources.	Understanding	Formative		
		SLO: C-11-F-11	Apply knowledge of energy sources and their properties to critically evaluate the advantages and disadvantages of different energy sources and make informed decisions about energy consumption.	Application	Formative		
							Total Lectures for theory = 150

G	Practically perform chemical experiments	SLO: C-11-G-01	Identify the chemical hazards in the lab in context of the experiment being conducted.	Knowledge	Formative	Question will not be asked in PBA, however, it will be part of Lab work	02
		SLO: C-11-G-02	Test that the equipment is working properly without any potential risk of injury before conducting an experiment.	Knowledge	Formative		
		SLO: C-11-G-03	Ensure that workspace for conducting the experiment is not crowded with apparatus as to be hazardous.	Knowledge	Formative		
		SLO: C-11-G-04	Ensure that safe distance is kept at all times from other investigators who may be handling lab apparatus.	Knowledge	Formative		
		SLO: C-11-G-05	Identify what potential bodily harm could occur from physical, chemical, biological and safety hazards in the context of the experiment being conducted.	Knowledge	Formative		
		SLO: C-11-G-06	Recognise that it is always better to ask for help from the lab instructor when unsure of how to use new apparatus.	Knowledge	Formative		
		SLO: C-11-G-07	Identify the proper waste disposal system for chemicals being used.	Knowledge	Formative		

	Practically perform chemical experiments	SLO: C-11-G-08	Set up apparatus following instructions given in written ordiagrammatic form.	Understanding	Formative	Question will not be asked in PBA, however, it will be part of Lab work	
		SLO: C-11-G-09	Use apparatus to collect an appropriate quantity of data	Understanding	Formative		
		SLO: C-11-G-10	Make observations, including subtle differences in colour, solubility or quantity of materials	Understanding	Formative		
		SLO: C-11-G-11	Make measurements using pipettes, burettes, measuring cylinders, thermometers and other common laboratory apparatus.	Understanding	Formative		
		SLO: C-11-G-12	Decide how many tests or observations to perform	Understanding	Formative		
		SLO: C-11-G-13	Identify where repeated readings or observations are appropriate	Understanding	Formative		
		SLO: C-11-G-14	Replicate readings or observations as necessary, including where an anomaly is suspected	Understanding	Formative		
		SLO: C-11-G-15	Identify where confirmatory tests are appropriate and the nature of such tests	Understanding	Formative		
		SLO: C-11-G-16	Select reagents to distinguish between given ions.	Knowledge	Formative		
		SLO: C-11-G-17	Carry out procedures using simple apparatus, in situations where the method may not be familiar to the candidate	Application	Formative		

	Acid Base Titration	SLO: C-11-G-18	Describe an acid–base titration to include the use of a: a. burette b. volumetric pipette c. suitable indicator	Understanding	Summative for Practical Based Assessment (PBA).	Laboratory work- will be assessed in PBA.	16
		SLO: C-11-G-19	Describe how to identify the endpoint of a titration using an indicator	Understanding	Summative for PBA.		
		SLO: C-11-G-20	Describe tests to identify the anions: a. CO_3^{2-} : By reaction with dilute acid and then testing for carbon dioxide gas. b. $\text{Cl}^- / \text{Br}^- / \text{I}^-$: By acidifying with dilute nitric acid then adding aqueous silver nitrate. c. NO_3^- : By reduction with aluminum foil and aqueous sodium hydroxide and then testing for ammonia gas. d. SO_4^{2-} : By acidifying with dilute nitric acid then adding aqueous barium nitrate. e. SO_3^{2-} : By reaction with acidified aqueous potassium manganate(VII).	Understanding	Summative for PBA.		
		SLO: C-11-G-21	Describe tests using aqueous sodium hydroxide and aqueous ammonia to identify the aqueous cations: a) aluminum, Al^{3+} b) ammonium, NH_4^{1+} c) calcium, Ca^{2+} d) chromium(III), Cr^{3+} e) copper(II), Cu^{2+} f) iron(II), Fe^{2+} g) iron(III), Fe^{3+} h) zinc, Zn^{2+}	Understanding	Summative for PBA.		
		SLO: C-11-G-22	Present numerical data, values or observations in a single table of results with headings and units that conform to accepted scientific conventions.	Application	Summative for PBA.		

	Data representation and analysis	SLO: C-11-G-22	Record raw readings of a quantity to the same degree of precision and observations to the same level of detail.	Application	Summative for PBA.		02
		SLO: C-11-G-23	Show working in calculations and key steps in reasoning.	Application	Summative for PBA.		
		SLO: C-11-G-24	Use the correct number of significant figures for calculated quantities.	Application	Summative for PBA.		
		SLO: C-11-G-25	Draw an appropriate table in advance of taking readings or making observations and record all data in the table.	Application	Summative for PBA.		
		SLO: C-11-G-26	Use the appropriate presentation method to produce a clear presentation of the data.	Application	Summative for PBA.		
		SLO: C-11-G-27	Plot appropriate variables on appropriate, clearly labelled x- and y-axes with carefully chosen scales.	Application	Summative for PBA.		
		SLO: C-11-G-28	Draw straight lines or smooth curves of best fit to show the trend of a graph.	Application	Summative for PBA.		
		SLO: C-11-G-29	Describe the patterns and trends shown by data in tables and graphs.	Application	Summative for PBA.		
		SLO: C-11-G-30	Describe and summarize the key points of a set of observations.	Application	Summative for PBA.		
		SLO: C-11-G-31	Determine the gradient of a straight-line graph and extrapolate the line of a graph.	Application	Summative for PBA.		
		SLO: C-11-G-32	Draw conclusions from an experiment, giving an outline description of the main features of the data, considering whether experimental data support a given hypothesis, and making further predictions.	Application	Summative for PBA.		

**Laboratory work-
will be assessed in
PBA.**

Suggest improvements based on validity, reliability and safety.	SLO: C-11-G-33	Draw conclusions from interpretations of observations, data and calculated values.	Application	Formative	Question will not be asked in PBA, however, it will be part of Lab work	02
	SLO: C-11-G-34	Make scientific explanations of data, observations and conclusions that they have described.	Application	Formative		
	SLO: C-11-G-35	Analyse intrinsic errors in measuring device.	Application	Formative		
	SLO: C-11-G-36	Describe systematic errors.	Understanding	Formative		
	SLO: C-11-G-37	Identify the most significant sources of error in an experiment	Understanding	Formative		
	SLO: C-11-G-38	State the uncertainty in a quantitative measurement and express such uncertainty in a measurement as an actual or percentage error	Understanding	Formative		
	SLO: C-11-G-39	Analyse the limitations of the experimental design and propose appropriate modifications that will improve the accuracy of the experiment.	Application	Formative		
	SLO: C-11-G-40	Evaluate the validity of the methods used	Understanding	Formative		
	SLO: C-11-G-41	Explain improvements or extensions to the methods used.	Understanding	Formative		
	SLO: C-11-G-42	Apply scientific language effectively.	Application	Formative		
	SLO: C-11-G-43	Document the work of others and sources of information used	Application	Formative		
	SLO: C-11-G-44	Suggest ways in which to extend the investigation to answer a new question.	Application	Formative		
	SLO: C-11-G-45	Suggest alternate chemicals in experimental design which contribute to green chemistry.	Application	Formative		
Total lectures for Practical's = 22 Total Lectures for Theory + practical's = 150 + 20 = 172						